Experiment 1

Aim : Implement Linear and Logistic Regression on real-world datasets

Data Source : (Used Car Price Prediction Dataset)

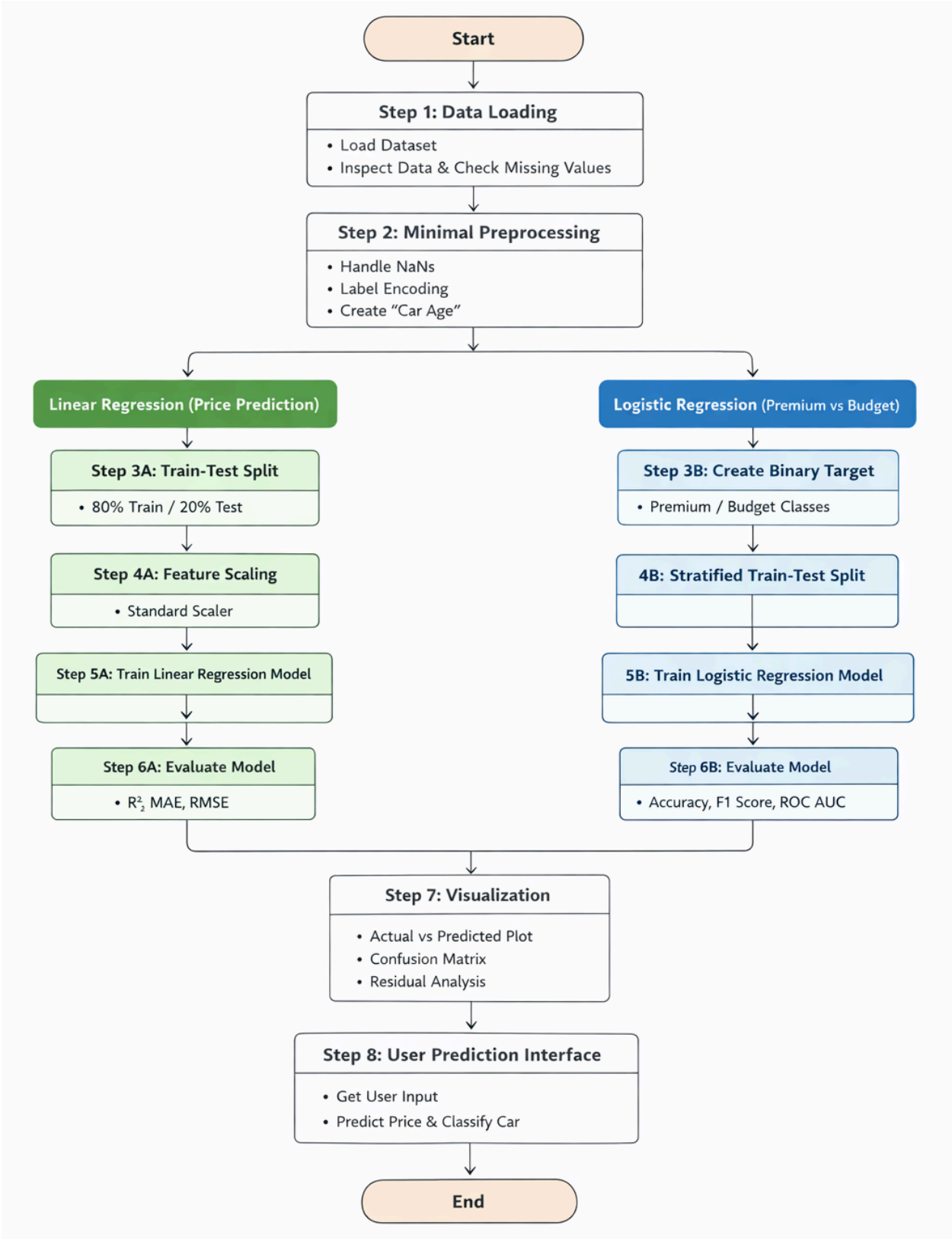
- Dataset Type: Structured tabular dataset
- File Format: CSV (Comma Separated Values)
- Total Records (Rows): 7,400
- Total Attributes (Columns): 29
- Domain: Used Car Sales / Automobile Marketplace Data
- Dataset Link

Data Description :
This dataset contains information about used cars listed for sale, including vehicle specifications, pricing details, advertisement information, and financial options.

Column Name	Description	Data Type
car_name	Name of the car	Object
yr_mfr	Year of manufacture	Integer
fuel_type	Type of fuel (Petrol/Diesel/etc.)	Object
kms_run	Kilometers driven	Integer
sale_price	Selling price of the car	Integer
city	City where car is listed	Object
times_viewed	Number of times listing was viewed	Integer
body_type	Car body type (Hatchback/Sedan/SUV)	Object
transmission	Manual/Automatic	Object
variant	Car variant/model version	Object
assured_buy	Whether platform assures purchase	Boolean
registered_city	City of vehicle registration	Object
registered_state	State of registration	Object
is_hot	Popular/trending listing indicator	Boolean
rto	RTO registration code	Object

source	Source of listing	Object
make	Car manufacturer brand	Object
model	Car model	Object
car_availability	Availability status	Object
total_owners	Number of previous owners	Integer
broker_quote	Broker suggested price	Integer
original_price	Original showroom price	Float
car_rating	Rating condition (Good/Great etc.)	Object
ad_created_on	Advertisement creation date	Object (Date)
fitness_certificate	Fitness certificate available	Boolean
emi_starts_from	EMI starting amount	Integer
booking_down_pymnt	Booking down payment amount	Integer
reserved	Whether car is reserved	Boolean
warranty_avail	Warranty availability	Boolean

Methodology Workflow :



Theory :

Regression techniques are fundamental supervised machine learning methods used to analyze relationships between variables and make predictions from real-world data. In this experiment, Linear Regression and Logistic Regression are implemented to understand both continuous prediction and binary classification problems using real-world datasets.

<u>Linear Regression :</u>

Linear Regression is a statistical learning technique used to model the relationship between a dependent (output) variable and one or more independent (input) variables by fitting a linear equation to observed data.

Mathematical Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon$$

Where:

- yyy is the predicted output
- x1,x2,…,xnx_1, x_2, \dots, x_nx1,x2,…,xn are input features
- β0\beta_0β0 is the intercept
- β1,β2,…,βn\beta_1, \beta_2, \dots, \beta_nβ1,β2,…,βn are model parameters
- ε\varepsilonε represents random error

Objective Function

The model parameters are estimated by minimizing the Mean Squared Error (MSE):

$$MSE = \frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

This ensures that the predicted values are as close as possible to the actual values.

Applications

Linear Regression is commonly used for:

- Price prediction
- Sales forecasting
- Trend analysis

<u>Logistic Regression :</u>

Logistic Regression is a classification algorithm used to predict outcomes where the dependent variable is binary, such as 0 or 1, true or false, or yes or no.

Logistic Model

$$P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_n x_n)}}$$

The model uses the sigmoid function to convert linear outputs into probability values between 0 and 1.

Cost Function

Logistic Regression minimizes the **log loss (binary cross-entropy)**:

$$J(\beta) = -\frac{1}{m}\sum[y\log(\hat{y}) + (1-y)\log(1-\hat{y})]$$

Applications

Logistic Regression is used in:

- Disease prediction
- Spam detection
- Fraud detection

Use of Real-World Datasets :

Real-world datasets often contain:

- Missing values
- Noise and outliers
- Different data scales

Therefore, preprocessing steps such as **data cleaning, feature scaling, and data splitting** are essential before model training. These steps improve model accuracy and generalization.

Model Evaluation Metrics :

Linear Regression Metrics

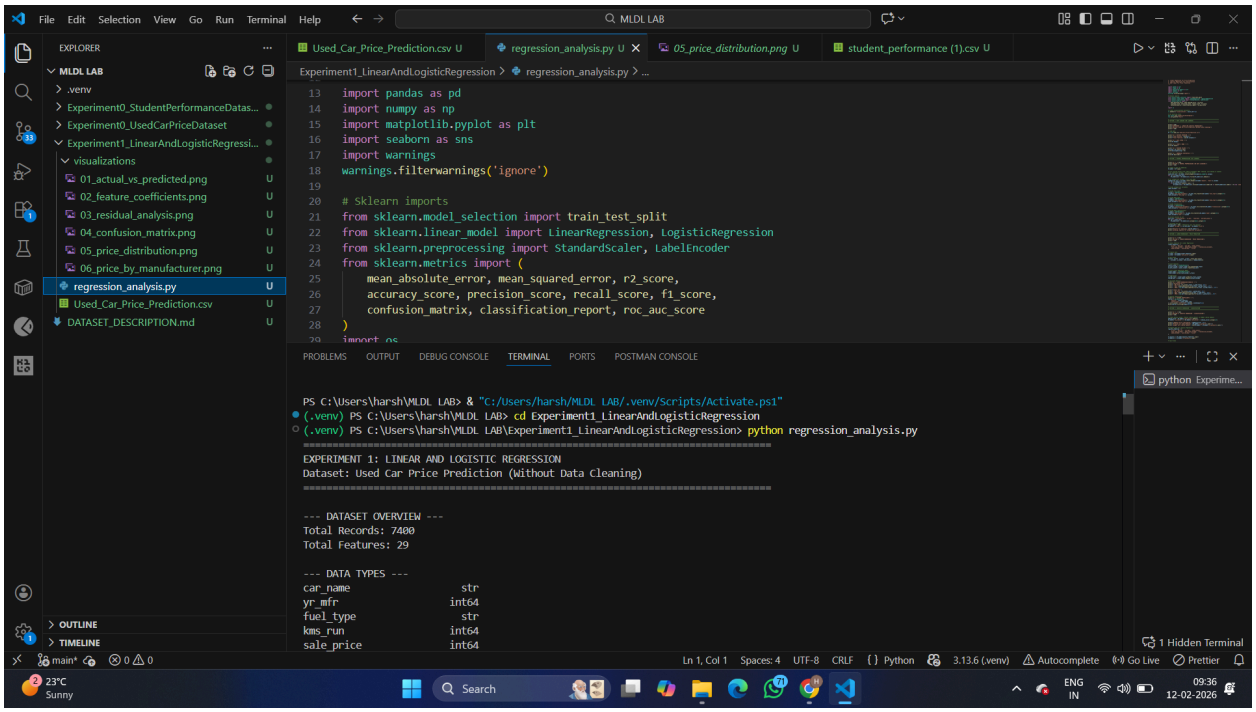
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

- R^2 Score

Logistic Regression Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Output :



```
(.venv) PS C:\Users\harsh\MLDL LAB\Experiment1_LinearAndLogisticRegression> python regression_analysis.py
3       maruti swift    2013  ...    False           False
4  hyundai grand i10    2015  ...    False           False

[5 rows x 29 columns]

--- MISSING VALUES ---
body_type              103
transmission           556
registered_city         10
registered_state        10
source                 126
car_availability       620
original_price        3280
car_rating               9
ad_created_on            1
fitness_certificate      8
dtype: int64

--- NUMERICAL STATISTICS ---
            yr_mfr       kms_run  ...  emi_starts_from  booking_down_pymnt
count  7400.000000    7400.000000  ...      7400.000000         7400.000000
mean   2013.885676   62624.520000  ...     10565.942027        68233.529054
std       3.087613   43532.042371  ...      6566.468434        42405.389468
min    1996.000000      60.000000  ...         0.000000            0.000000
25%    2012.000000   31885.250000  ...      6531.250000        42176.250000
50%    2014.000000   55765.000000  ...      8883.000000        57367.500000
75%    2016.000000   84184.000000  ...     12546.250000        81022.500000
max    2021.000000  996609.000000  ...     89798.000000       579901.000000

[8 rows x 9 columns]

================================================================================
SECTION 2: MINIMAL PREPROCESSING (NO DATA CLEANING)
================================================================================
```

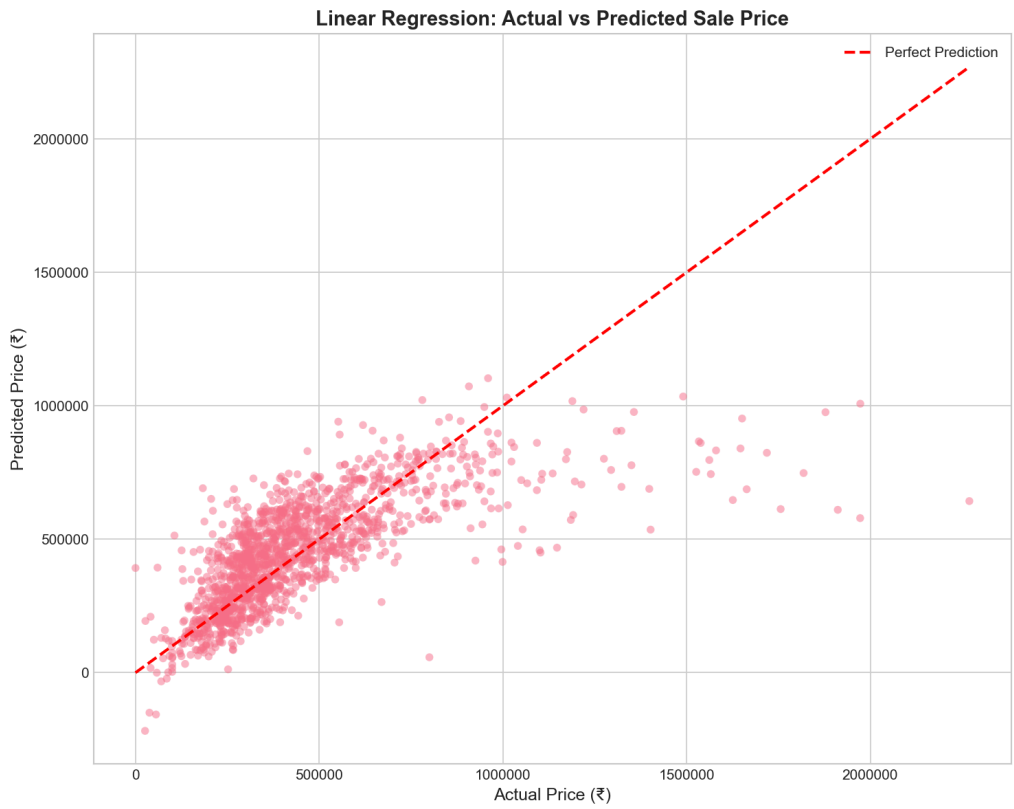
```
(.venv) PS C:\Users\harsh\MLDL LAB\Experiment1_LinearAndLogisticRegression> python regression_analysis.py

================================================================================
SECTION 2: MINIMAL PREPROCESSING (NO DATA CLEANING)
================================================================================
Records for modeling: 7400
Encoding complete for categorical variables

================================================================================
SECTION 3: LINEAR REGRESSION - PRICE PREDICTION
================================================================================

--- LINEAR REGRESSION RESULTS ---

Training Set:
  R² Score: 0.4568
  MAE: ₹128,980.76
  RMSE: ₹211,607.45

Test Set:
  R² Score: 0.5021
  MAE: ₹122,496.59
  RMSE: ₹186,179.15

--- FEATURE COEFFICIENTS ---
              Feature   Coefficient
   fuel_type_encoded -76552.232697
              yr_mfr  61269.750387
             car_age -61269.750387
transmission_encoded -59010.051708
   body_type_encoded  58901.138278
             kms_run -13547.973171
         assured_buy -11321.279177
        make_encoded  -6454.914048
              is_hot   2685.685070
```

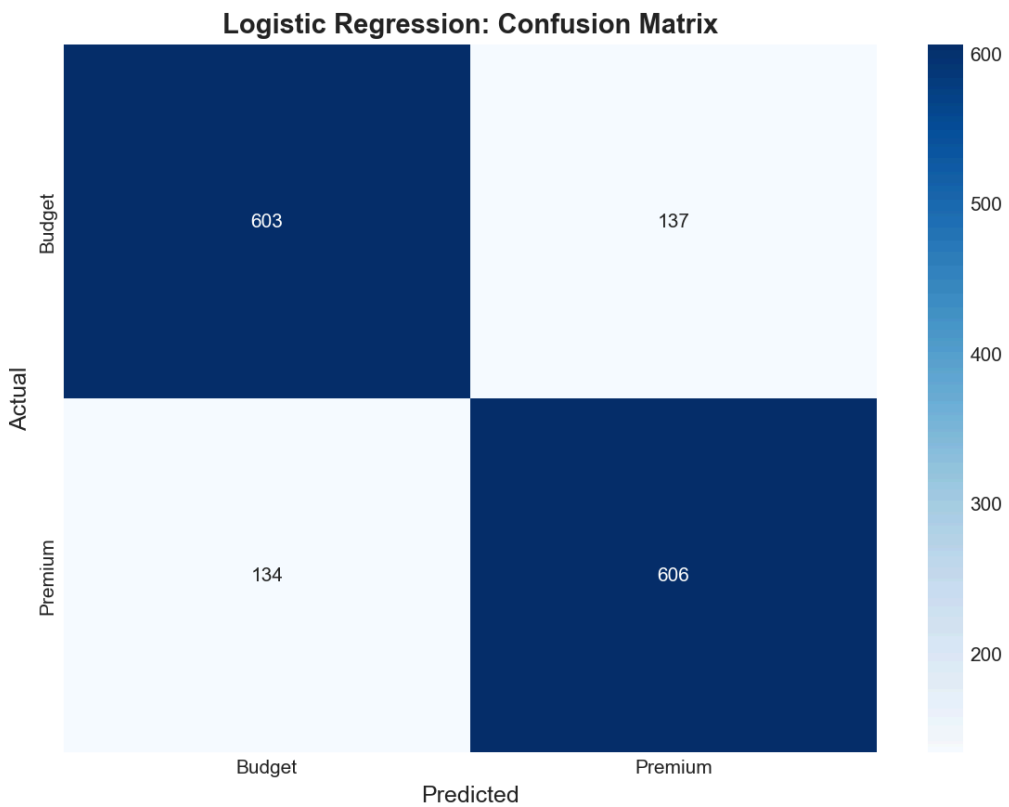
```
(.venv) PS C:\Users\harsh\MLDL LAB\Experiment1_LinearAndLogisticRegression> python regression_analysis.py
================================================================================
SECTION 4: LOGISTIC REGRESSION - CLASSIFICATION
================================================================================

Median Price (Threshold): ₹382,449.00
Premium Cars (Above Median): 3700
Budget Cars (Below Median): 3700

--- LOGISTIC REGRESSION RESULTS ---

Classification Metrics:
  Accuracy:  0.8169
  Precision: 0.8156
  Recall:    0.8189
  F1 Score:  0.8173
  ROC AUC:   0.9036

--- CONFUSION MATRIX ---
                Predicted
               Budget  Premium
Actual Budget     603      137
Actual Premium    134      606

--- CLASSIFICATION REPORT ---
              precision    recall  f1-score   support

      Budget       0.82      0.81      0.82       740
     Premium       0.82      0.82      0.82       740

    accuracy                           0.82      1480
   macro avg       0.82      0.82      0.82      1480
weighted avg       0.82      0.82      0.82      1480
```

```
(.venv) PS C:\Users\harsh\MLDL LAB\Experiment1_LinearAndLogisticRegression> python regression_analysis.py

================================================================================
SECTION 5: GENERATING VISUALIZATIONS
================================================================================
Creating Visualization 1: Actual vs Predicted...
Creating Visualization 2: Feature Coefficients...
Creating Visualization 3: Residual Plot...
Creating Visualization 4: Confusion Matrix...
Creating Visualization 5: Price Distribution...
Creating Visualization 6: Price by Manufacturer...

All visualizations saved to 'visualizations/' folder!

================================================================================
SECTION 6: USER INPUT FOR PRICE PREDICTION
================================================================================

================================================================================
SECTION 7: SUMMARY
================================================================================

--- MODEL PERFORMANCE SUMMARY ---

📈 LINEAR REGRESSION (Price Prediction):
   • R² Score (Test): 0.5021
   • MAE: ₹122,496.59
   • RMSE: ₹186,179.15

📊 LOGISTIC REGRESSION (Premium/Budget Classification):
   • Accuracy: 0.8169
   • F1 Score: 0.8173
   • ROC AUC: 0.9036

--- FILES GENERATED ---
```

```
(.venv) PS C:\Users\harsh\MLDL LAB\Experiment1_LinearAndLogisticRegression> python regression_analysis.py

======================================================================
INTERACTIVE PRICE PREDICTION
======================================================================

--- OPTIONS ---
1. Predict Car Price (Enter your car details)

Enter your choice (1 or 2): 1

------------------------------------------------------------
USED CAR PRICE PREDICTION SYSTEM
------------------------------------------------------------

--- Available Options ---
Fuel Types: ['diesel', 'electric', 'petrol', 'petrol & cng', 'petrol & lpg']
Body Types: ['hatchback', 'luxury sedan', 'luxury suv', 'sedan', 'suv']
Transmission: ['automatic', 'manual']
Manufacturers: ['audi', 'bmw', 'chevrolet', 'datsun', 'fiat', 'ford', 'honda', 'hyundai', 'isuzu', 'jaguar', 'jeep', 'kia', 'mahindra', 'mah
indra renault', 'maruti', 'mercedes benz', 'mg', 'mitsubishi', 'nissan', 'opel', 'renault', 'skoda', 'ssangyong', 'tata', 'toyota', 'volkswa
gen', 'volvo']

--- Enter Car Details ---
Year of Manufacture (e.g., 2015): 2010
Kilometers Run (e.g., 50000): 20000
Total Owners (e.g., 1, 2, 3): 2

Fuel Types: ['diesel', 'electric', 'petrol', 'petrol & cng', 'petrol & lpg']
Fuel Type: petrol

Body Types: ['hatchback', 'luxury sedan', 'luxury suv', 'sedan', 'suv']
Body Type: suv

Transmission: ['automatic', 'manual']
```

```
(.venv) PS C:\Users\harsh\MLDL LAB\Experiment1_LinearAndLogisticRegression> python regression_analysis.py
indra renault', 'maruti', 'mercedes benz', 'mg', 'mitsubishi', 'nissan', 'opel', 'renault', 'skoda', 'ssangyong', 'tata', 'toyota', 'volkswa
gen', 'volvo']
Manufacturer: audi
Assured Buy? (yes/no): y
Is Hot Deal? (yes/no): y

============================================================
PREDICTION RESULTS
============================================================

🚗 Car Details Summary:
   • Year: 2010
   • Kilometers: 20,000
   • Car Age: 11 years
   • Owners: 2
   • Fuel: petrol
   • Body: suv
   • Transmission: manual
   • Manufacturer: audi

💰 PREDICTED SALE PRICE: ₹384,811.91

📊 Category: BUDGET (Confidence: 22.0%)
   (Threshold: ₹382,449.00)

📈 Estimated Price Range:
   • Lower: ₹262,315.32
   • Upper: ₹507,308.51

--- OPTIONS ---
1. Predict Car Price (Enter your car details)
2. Exit

Enter your choice (1 or 2):
```


Linear Regression: Actual vs Predicted Sale Price
Perfect Prediction
Predicted Price (₹)
Actual Price (₹)
0
500000
1000000
1500000
2000000


Logistic Regression: Confusion Matrix
Actual
Predicted
Budget
Premium
603
137
134
606
600
500
400
300
200

Conclusion :

Linear Regression and Logistic Regression are effective and widely used machine learning techniques for prediction and classification tasks. When applied to real-world datasets with proper preprocessing and evaluation, they provide accurate, interpretable, and reliable results, making them essential tools in data analysis and decision-making.